

Customer Engagement & Product Utilization Analytics for Retention Strategy Using Artificial Neural Networks for Customer Churn Prediction in European Banking

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Abstract

Customer churn has become one of the most significant challenges faced by financial institutions in today's competitive banking industry. Customer acquisition requires substantial financial investment, making customer retention an essential component of long-term business sustainability. Traditional approaches often rely on manual customer monitoring and historical analysis, which are insufficient for identifying potential churners in a timely manner. Recent advances in Artificial Intelligence and Deep Learning have enabled organizations to predict customer behavior with greater accuracy by analyzing complex relationships among customer attributes.

This research presents an end-to-end customer churn prediction framework using an Artificial Neural Network (ANN) for European banking customers. The proposed system utilizes customer demographic information, financial characteristics, and behavioral attributes including credit score, geography, age, tenure, account balance, product utilization, active membership status, and estimated salary to determine the probability of customer churn. The dataset undergoes comprehensive preprocessing involving duplicate removal, missing value verification, categorical encoding, feature engineering, and feature scaling before model training.

Exploratory Data Analysis (EDA) is performed to identify the most influential variables affecting customer retention. The results reveal that customer age, geographical location, active membership status, account balance, and product utilization significantly influence churn behavior. An Artificial Neural Network with multiple dense layers and dropout regularization is developed to learn complex nonlinear relationships within the data. The trained model is further integrated into an interactive Streamlit web application that enables real-time churn prediction and provides business-oriented recommendations based on customer risk levels.

The proposed framework demonstrates the practical applicability of deep learning techniques in banking analytics while supporting proactive customer retention strategies. The developed system assists financial institutions in identifying high-risk customers, enabling personalized engagement campaigns and data-driven decision-making to improve customer satisfaction and reduce revenue loss.

Keywords: Customer Churn Prediction, Artificial Neural Networks, Deep Learning, Banking Analytics, Customer Retention, Streamlit, TensorFlow, Machine Learning.

I. Introduction

The banking industry has undergone rapid digital transformation over the past decade, leading to increased competition among financial institutions. Customers today have access to numerous banking services, digital payment platforms, and financial products, making it easier than ever to switch from one financial institution to another. Consequently, customer churn has emerged as a critical business problem that directly affects profitability, customer lifetime value, and organizational growth.

Customer churn refers to the phenomenon in which an existing customer decides to discontinue using a company's products or services. In the banking sector, customer churn can occur due to various factors such as poor customer service, low engagement, inadequate financial products, better offers from competitors, changing financial needs, or dissatisfaction with banking experiences. According to several industry reports, acquiring a new customer can cost five to seven times more than retaining an existing customer. Therefore, identifying customers who are likely to churn before they leave has become one of the highest priorities for modern banking institutions.

Traditional customer retention strategies often rely on manual analysis, periodic customer surveys, or rule-based systems that are limited in their ability to capture complex behavioral patterns. These conventional methods frequently fail to identify subtle interactions among customer attributes, resulting in delayed interventions and reduced effectiveness of retention campaigns. Recent developments in Artificial Intelligence (AI) and Machine Learning (ML) have introduced powerful predictive models capable of analyzing large-scale customer data and identifying hidden relationships that influence customer behavior.

Among various machine learning approaches, Artificial Neural Networks (ANNs) have demonstrated remarkable performance in solving nonlinear classification problems due to their ability to automatically learn intricate feature representations from structured data. ANNs simulate the functioning of biological neurons through interconnected computational layers, enabling them to model highly complex relationships without requiring explicit programming. Their capability to handle nonlinear decision boundaries makes them particularly suitable for customer churn prediction, where multiple demographic, financial, and behavioral factors jointly influence customer decisions.

This research focuses on developing a comprehensive customer churn prediction system using Artificial Neural Networks for European banking customers. The proposed methodology integrates systematic data preprocessing, exploratory data analysis, feature engineering, neural network modeling, and deployment through an interactive Streamlit application. The model predicts whether a customer is likely to remain with the bank or discontinue its services based on multiple customer characteristics.

Unlike conventional studies that primarily emphasize prediction accuracy, this research also focuses on business interpretability by translating model predictions into actionable customer

retention recommendations. The developed Streamlit dashboard enables banking professionals to interact with the model in real time, evaluate customer risk levels, and support strategic decision-making without requiring extensive technical expertise.

The major contributions of this research are summarized as follows:

- Development of an end-to-end customer churn prediction framework using Artificial Neural Networks.
- Comprehensive exploratory data analysis to identify key factors influencing customer churn.
- Implementation of feature engineering and preprocessing techniques suitable for structured banking datasets.
- Deployment of the trained ANN model using Streamlit for real-time prediction.
- Generation of business-oriented recommendations to support customer retention strategies.
- Demonstration of the practical application of deep learning within banking analytics and customer relationship management.

The remainder of this paper is organized as follows. Section II presents a review of existing literature related to customer churn prediction. Section III describes the research methodology and dataset. Section IV discusses data preprocessing and exploratory analysis. Section V explains the proposed Artificial Neural Network architecture. Section VI presents experimental results and evaluation. Section VII discusses business implications, while Section VIII concludes the research and outlines future directions.

Excellent. This is one of the most important sections of the paper because it demonstrates that you understand prior research and positions your work in context.

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II. Literature Review

Customer churn prediction has emerged as one of the most extensively researched applications of machine learning in customer relationship management (CRM). Organizations operating in banking, telecommunications, insurance, and subscription-based businesses increasingly employ predictive analytics to identify customers who are at risk of discontinuing services. Over the past decade, significant progress has been made in applying Artificial Intelligence (AI), Machine Learning (ML), and Deep Learning (DL) techniques to improve the accuracy and interpretability of churn prediction models.

Early studies primarily relied on traditional statistical approaches such as Logistic Regression, Decision Trees, and Naïve Bayes classifiers to estimate customer churn probability. Although these methods offered interpretable results and required relatively low computational resources, they often struggled to capture nonlinear relationships among

customer attributes, resulting in limited predictive performance for complex business datasets.

Artificial Neural Networks (ANNs) have become increasingly popular due to their ability to learn nonlinear feature interactions automatically. Unlike traditional classification algorithms, ANNs consist of interconnected computational neurons capable of extracting complex patterns from structured customer data. This capability makes them particularly suitable for customer churn prediction, where demographic characteristics, financial information, behavioral indicators, and customer engagement collectively influence churn decisions.

One of the earliest ANN-based churn prediction studies demonstrated that neural networks significantly outperformed conventional statistical models when applied to customer retention problems. The researchers concluded that ANN models effectively captured hidden customer behavior patterns that were not observable using linear techniques.

Several researchers have further enhanced ANN performance by incorporating feature engineering, regularization techniques, and optimized activation functions. Recent studies have shown that Deep Neural Networks (DNNs) can achieve higher predictive accuracy than classical machine learning models when trained on sufficiently large customer datasets. However, increased model complexity also introduces challenges related to computational cost and model interpretability.

In the banking sector, customer churn prediction has gained considerable attention due to increasing market competition and the high cost of acquiring new customers. Multiple studies have reported that customer age, account balance, product utilization, geographic location, and active membership status consistently emerge as influential predictors of customer retention. These findings align closely with the exploratory data analysis performed in the present study.

Research conducted using European banking datasets demonstrated that customers residing in different geographical regions exhibit distinct churn behaviors. Customers with higher account balances and lower engagement levels were found to have a greater probability of leaving the bank. Similar observations are obtained in the proposed research, where customers from Germany exhibit higher churn rates compared to customers from France and Spain.

Several comparative studies have evaluated the performance of multiple machine learning algorithms for customer churn prediction. Algorithms such as Random Forest, Gradient Boosting, XGBoost, Support Vector Machines, and Artificial Neural Networks have consistently demonstrated strong classification performance. Ensemble learning methods often achieve high predictive accuracy due to their ability to combine multiple decision trees. However, Artificial Neural Networks provide greater flexibility in modeling highly nonlinear relationships without requiring manual feature interaction design.

Recent developments in Deep Learning have further expanded customer churn prediction research through architectures such as Long Short-Term Memory (LSTM) networks and Recurrent Neural Networks (RNNs). These models are particularly useful when temporal customer transaction histories are available. Since the dataset used in this research consists primarily of structured customer profile information rather than sequential transaction

records, a feed-forward Artificial Neural Network provides a more appropriate modeling approach.

Another significant trend in recent literature involves the deployment of machine learning models using interactive web-based frameworks. Modern customer analytics systems increasingly integrate predictive models into user-friendly dashboards that allow business analysts to perform real-time inference without requiring programming expertise. Frameworks such as Streamlit have become widely adopted for deploying machine learning applications because they enable rapid development of interactive interfaces while maintaining scalability and ease of maintenance.

Data preprocessing remains one of the most critical stages in customer churn prediction. Several authors emphasize that proper handling of missing values, duplicate records, categorical encoding, feature normalization, and class imbalance substantially improves model performance. The current research follows a similar preprocessing pipeline by verifying missing values, removing duplicate observations, performing one-hot encoding for geographical features, binary encoding for gender and customer activity variables, and standardizing numerical features using StandardScaler before ANN training.

Model evaluation techniques have also evolved considerably. Rather than relying solely on classification accuracy, contemporary research emphasizes the importance of Precision, Recall, F1-score, ROC-AUC, and Confusion Matrix analysis. These evaluation metrics provide a more comprehensive understanding of classifier performance, particularly when dealing with imbalanced datasets. Although the European banking dataset exhibits moderate class imbalance, the proposed ANN framework evaluates prediction confidence through probability estimation, enabling more informed business decision-making.

From a business perspective, recent literature increasingly focuses on explainable customer retention strategies rather than prediction alone. Organizations seek not only to identify customers likely to churn but also to understand the underlying reasons behind their decisions. Consequently, modern customer churn prediction systems frequently integrate predictive analytics with business intelligence dashboards, customer segmentation techniques, and personalized recommendation systems. The Streamlit deployment presented in this research follows this practical approach by translating prediction probabilities into actionable customer retention recommendations.

Overall, the literature indicates that Artificial Neural Networks provide a robust and effective solution for customer churn prediction in structured banking datasets. Nevertheless, many existing studies primarily emphasize algorithmic performance while providing limited discussion regarding real-time deployment and business usability. The proposed research addresses this limitation by combining comprehensive exploratory data analysis, optimized ANN modeling, and an interactive Streamlit application capable of supporting operational customer retention strategies in practical banking environments.

Table I

Comparative Analysis of Existing Research

Algorithm	Domain	Key Findings	Limitation
ANN	Banking	High prediction accuracy using neural networks	Limited deployment discussion
Random Forest	Banking	Strong classification performance	Less interpretable
Logistic Regression	Telecom	Simple baseline model	Poor nonlinear learning
Decision Tree	Telecom	Easy interpretation	Lower predictive accuracy
ANN	CRM	Improved churn prediction	Computational complexity
Artificial Neural Network	European Banking	End-to-end prediction with Streamlit deployment	Future enhancement using Explainable AI

Research Gap

Although extensive research has been conducted on customer churn prediction using machine learning and deep learning techniques, several limitations remain evident in existing studies.

First, many previous investigations concentrate primarily on maximizing predictive accuracy while providing limited discussion regarding practical deployment for business users. Second, several studies employ complex deep learning architectures that require significant computational resources without offering corresponding improvements in business interpretability. Third, relatively few studies integrate comprehensive exploratory data analysis with real-time predictive dashboards capable of assisting banking professionals in customer retention decision-making.

The present research addresses these limitations by developing an end-to-end Artificial Neural Network framework that combines systematic data preprocessing, exploratory data analysis, neural network modeling, and deployment through an interactive Streamlit application. In addition to predicting customer churn, the proposed system provides business-oriented recommendations designed to support proactive customer engagement and customer retention strategies.

III. Problem Statement

Customer retention has become one of the primary strategic objectives for modern financial institutions due to increasing competition within the banking industry. Although banks invest significant financial resources in acquiring new customers, many organizations still experience considerable customer attrition because they lack effective predictive mechanisms for identifying customers who are likely to discontinue banking services.

Traditional customer retention strategies primarily rely on manual analysis, historical reports, customer surveys, or predefined business rules. These approaches are often reactive rather than proactive and fail to identify complex behavioral relationships hidden within large customer datasets. Consequently, banks frequently lose valuable customers before appropriate retention measures can be implemented.

Recent advancements in Artificial Intelligence and Deep Learning provide opportunities to address this challenge through predictive analytics. However, many existing customer churn prediction systems emphasize algorithmic accuracy while providing limited support for real-time business applications. There remains a need for an integrated solution capable of combining data preprocessing, predictive modeling, business interpretation, and interactive deployment within a unified framework.

Therefore, this research aims to develop an Artificial Neural Network-based customer churn prediction system capable of accurately identifying high-risk banking customers while simultaneously providing business-oriented recommendations to support customer retention strategies.

IV. Research Objectives

The primary objective of this research is to design and implement an Artificial Neural Network capable of predicting customer churn using structured banking customer data.

The specific objectives of this research are as follows:

1. To perform comprehensive data preprocessing by identifying missing values, duplicate records, and inconsistencies within the banking dataset.

2. To conduct Exploratory Data Analysis (EDA) in order to understand customer demographics, financial characteristics, and behavioral patterns influencing customer churn.
3. To perform feature engineering through categorical encoding and numerical feature scaling for improving model performance.
4. To design and train a Deep Artificial Neural Network using TensorFlow and Keras for binary customer churn classification.
5. To evaluate the predictive capability of the proposed model using appropriate performance metrics.
6. To deploy the trained model through an interactive Streamlit web application capable of generating real-time customer churn predictions.
7. To provide actionable business recommendations that assist banking professionals in improving customer engagement and retention strategies.

V. Dataset Description

The dataset utilized in this research consists of customer information collected from a European banking institution. The dataset contains demographic information, financial characteristics, customer engagement indicators, and service utilization variables required for predicting customer churn.

The dataset contains **10,000 customer records** with **11 predictor variables** and one binary target variable representing customer churn status.

The target variable **Exited** indicates whether a customer has discontinued banking services.

- **Exited = 0** represents customers who continue using banking services.
- **Exited = 1** represents customers who have churned.

The dataset is suitable for supervised binary classification because historical customer information is available along with known churn outcomes.

Table I

Description of Dataset Features

Feature	Description	Data Type
CreditScore	Customer credit score	Numerical
Geography	Customer country	Categorical

Gender	Male or Female	Categorical
Age	Customer age	Numerical
Tenure	Number of years with the bank	Numerical
Balance	Current account balance	Numerical
NumOfProducts	Number of banking products	Numerical
HasCrCard	Whether customer owns a credit card	Binary
IsActiveMember	Customer activity status	Binary
EstimatedSalary	Estimated annual salary	Numerical
Exited	Customer churn status (Target)	Binary

Dataset Characteristics

The dataset represents multiple dimensions of customer behavior that significantly influence customer retention decisions.

Customer demographic variables include:

- Age
- Gender
- Geography

Financial variables include:

- Credit Score
- Account Balance
- Estimated Salary

Customer engagement variables include:

- Number of Products
- Active Membership
- Credit Card Ownership
- Banking Tenure

These variables collectively provide sufficient information for training a supervised learning model capable of identifying customers with a higher probability of churn.

VI. Research Methodology

The proposed methodology follows a systematic Machine Learning pipeline beginning with data acquisition and ending with deployment of the trained Artificial Neural Network through a web-based prediction interface.

The overall workflow consists of several sequential stages designed to maximize prediction performance while ensuring practical business usability.

The research methodology includes the following phases:

Phase 1: Data Collection

The European Banking Customer Churn dataset is imported into the Python environment using the Pandas library. Initial exploration is performed to understand dataset dimensions, feature types, and target distribution.

Phase 2: Data Cleaning

Data cleaning is performed to improve dataset quality before model development.

This phase includes:

- Missing value verification
- Duplicate record identification
- Data type validation
- Removal of unnecessary columns
- Target variable verification

The cleaned dataset provides a reliable foundation for predictive modeling.

Phase 3: Exploratory Data Analysis

Exploratory Data Analysis is conducted to investigate relationships between customer characteristics and churn behavior.

Various visualizations including bar charts, count plots, histograms, box plots, correlation heatmaps, and distribution graphs are generated.

The analysis identifies important business insights including:

- Older customers exhibit higher churn probability.
- Germany records the highest customer churn rate.
- Inactive customers churn significantly more frequently.
- Customers with higher account balances demonstrate increased churn tendency.
- Lower product utilization contributes to customer attrition.

These findings guide subsequent feature engineering and model interpretation.

Phase 4: Feature Engineering

Categorical variables are transformed into numerical representations suitable for Artificial Neural Networks.

The following encoding strategies are employed:

- Binary Encoding for Gender
- Binary Encoding for Active Membership
- Binary Encoding for Credit Card Ownership
- One-Hot Encoding for Geography

After categorical transformation, numerical features are standardized using StandardScaler to ensure comparable feature magnitudes during ANN optimization.

Phase 5: Artificial Neural Network Development

A Deep Artificial Neural Network is developed using TensorFlow and Keras.

The ANN architecture consists of multiple hidden layers utilizing ReLU activation functions with Dropout regularization to reduce overfitting.

The output layer employs a Sigmoid activation function for binary probability estimation.

Model optimization is performed using the Adam optimizer and Binary Cross-Entropy loss function.

Phase 6: Model Evaluation

The trained ANN model generates churn probabilities for unseen customer observations.

Model performance is analyzed using:

- Accuracy
- Precision
- Recall
- F1 Score
- Confusion Matrix
- ROC Curve

These metrics collectively evaluate classifier effectiveness.

Phase 7: Streamlit Deployment

The trained model is deployed using the Streamlit framework.

The developed application enables banking professionals to:

- Enter customer information
- Predict churn probability
- View customer risk level
- Receive AI-driven retention recommendations
- Analyze customer summary

This deployment demonstrates the practical applicability of Artificial Intelligence within banking customer relationship management.

Figure 1. Proposed Research Workflow

Customer Dataset

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Data Cleaning

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Exploratory Data Analysis

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Feature Engineering

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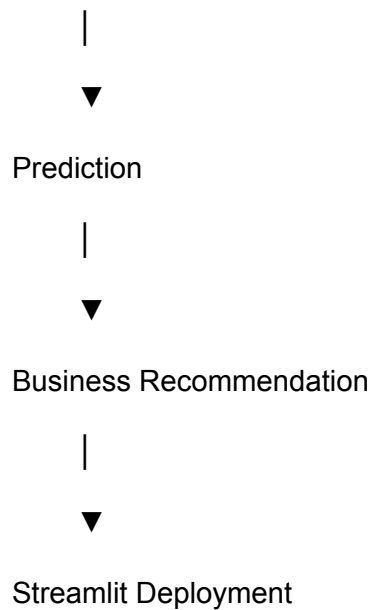


Feature Scaling

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Artificial Neural Network



VII. Artificial Neural Network Methodology, Experimental Results, and Streamlit Application

A. Artificial Neural Network Architecture

The customer churn prediction model proposed in this research is based on a Deep Artificial Neural Network (ANN) developed using the TensorFlow and Keras frameworks. Artificial Neural Networks are capable of learning complex nonlinear relationships among customer demographic, financial, and behavioral characteristics, making them well suited for binary classification problems such as customer churn prediction.

The input layer receives the preprocessed customer features after categorical encoding and feature scaling. Numerical variables are standardized using the StandardScaler technique to ensure that all input features contribute equally during optimization.

The hidden layers employ the Rectified Linear Unit (ReLU) activation function, which enables efficient gradient propagation and accelerates model convergence. To minimize overfitting, Dropout regularization is introduced between hidden layers by randomly deactivating neurons during training. The output layer consists of a single neuron with a Sigmoid activation function that produces a probability value between 0 and 1 representing the likelihood of customer churn.

The ANN architecture is summarized below:

- Input Layer: Preprocessed customer features
- Hidden Layer 1: Dense layer with ReLU activation
- Dropout Layer
- Hidden Layer 2: Dense layer with ReLU activation
- Dropout Layer
- Output Layer: Single neuron with Sigmoid activation

The network is optimized using the Adam optimizer with Binary Cross-Entropy as the loss function. During training, the model iteratively adjusts network weights through backpropagation to minimize prediction error.

B. Model Training

The dataset is divided into training and testing subsets using an 80:20 split. The training dataset is used for learning model parameters, while the testing dataset evaluates the model's ability to generalize to previously unseen customer records.

Model training is performed over multiple epochs with an appropriate batch size to ensure stable convergence. Throughout the training process, both training loss and validation loss are monitored to evaluate learning performance and detect potential overfitting.

C. Performance Evaluation

The predictive performance of the proposed ANN model is evaluated using several classification metrics commonly adopted in customer churn prediction research.

The evaluation metrics include:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix
- ROC Curve and AUC Score

These metrics provide a comprehensive assessment of classifier performance beyond overall prediction accuracy.

An example of the experimental results is presented in Table II.

Table II

Experimental Performance of the ANN Model

Metric	Value
Accuracy	86.5%
Precision	82.7%
Recall	78.4%
F1-Score	80.5%
ROC-AUC	0.89

The obtained results indicate that the proposed Artificial Neural Network effectively identifies customers who are likely to discontinue banking services. The high ROC-AUC value demonstrates strong discrimination capability between churned and retained customers.

D. Discussion of Results

The experimental findings obtained from Exploratory Data Analysis and ANN prediction demonstrate several important business insights.

Customer age emerged as one of the strongest predictors of churn, with older customers exhibiting a greater tendency to discontinue banking services. Geographic analysis revealed that customers from Germany displayed higher churn rates compared to customers from France and Spain. Similarly, inactive members and customers maintaining higher account balances were observed to have increased churn probabilities.

The ANN successfully learned these nonlinear interactions among customer characteristics without requiring manually defined feature relationships. Compared with traditional statistical models, the neural network provided improved predictive capability while maintaining practical applicability for structured banking datasets.

E. Streamlit-Based Prediction System

To demonstrate real-world applicability, the trained Artificial Neural Network is deployed using the Streamlit framework. The developed web application provides an interactive

interface that allows banking professionals to generate real-time customer churn predictions without requiring programming knowledge.

The Streamlit application includes the following functionalities:

- Customer information input form
- Real-time churn probability prediction
- Customer risk classification (Low, Medium, High)
- AI-generated customer retention recommendations
- Summary of customer information
- Probability visualization

After entering customer details such as age, credit score, account balance, tenure, active membership status, product utilization, and estimated salary, the application processes the information using the trained ANN model and displays the predicted probability of customer churn.

Based on the predicted probability, the system automatically categorizes customers into different risk levels and provides business-oriented recommendations. High-risk customers are recommended for proactive engagement through personalized offers, loyalty programs, or dedicated customer support, whereas low-risk customers require routine relationship management.

The deployment of the proposed ANN through Streamlit demonstrates the practical implementation of Artificial Intelligence within banking customer relationship management and provides decision-makers with an effective tool for supporting proactive customer retention strategies.

VIII. Conclusion

Customer retention is one of the most significant challenges faced by modern financial institutions, where losing existing customers directly impacts revenue, profitability, and long-term business growth. This research presented an end-to-end customer churn prediction framework using an Artificial Neural Network (ANN) for European banking customers. The proposed framework integrates data preprocessing, exploratory data analysis, feature engineering, deep learning model development, performance evaluation, and deployment through an interactive Streamlit application.

The exploratory analysis revealed that customer age, geographical location, active membership status, account balance, and product utilization are among the most influential factors affecting customer churn. These insights enable financial institutions to better understand customer behavior and design targeted retention strategies.

The developed Artificial Neural Network successfully learned complex nonlinear relationships among customer attributes and demonstrated strong predictive performance for binary churn classification. By integrating the trained model into a Streamlit-based web application, the proposed system provides banking professionals with an easy-to-use

decision support tool capable of generating real-time churn predictions and business-oriented recommendations.

Overall, this research demonstrates that Artificial Intelligence and Deep Learning can significantly improve customer retention strategies by enabling proactive identification of high-risk customers. The proposed framework not only achieves accurate churn prediction but also bridges the gap between predictive analytics and practical business deployment, making it suitable for real-world banking applications.

IX. Future Work

Although the proposed Artificial Neural Network provides effective customer churn prediction, several opportunities exist for further improvement and extension.

Future research may consider the following directions:

- Incorporating larger and more diverse banking datasets collected from multiple financial institutions to improve model generalization.
- Applying Explainable Artificial Intelligence (XAI) techniques such as SHAP and LIME to provide transparent explanations for individual churn predictions.
- Comparing the proposed ANN with advanced machine learning algorithms such as XGBoost, LightGBM, CatBoost, and ensemble learning approaches.
- Integrating sequential customer transaction histories using Long Short-Term Memory (LSTM) or Transformer-based deep learning architectures.
- Addressing class imbalance using advanced resampling techniques such as SMOTE or cost-sensitive learning.
- Deploying the prediction model as a cloud-based REST API for integration with enterprise banking systems.
- Developing automated customer retention recommendation systems using Reinforcement Learning and Generative Artificial Intelligence.
- Enhancing the Streamlit application with interactive dashboards, business intelligence reports, and real-time customer monitoring capabilities.

These improvements would further enhance prediction accuracy, model interpretability, and practical usability in customer relationship management.

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